Rear Wing Multi-Level Inverted Airfoil

CFD Analysis (SA vs k-ω SST)

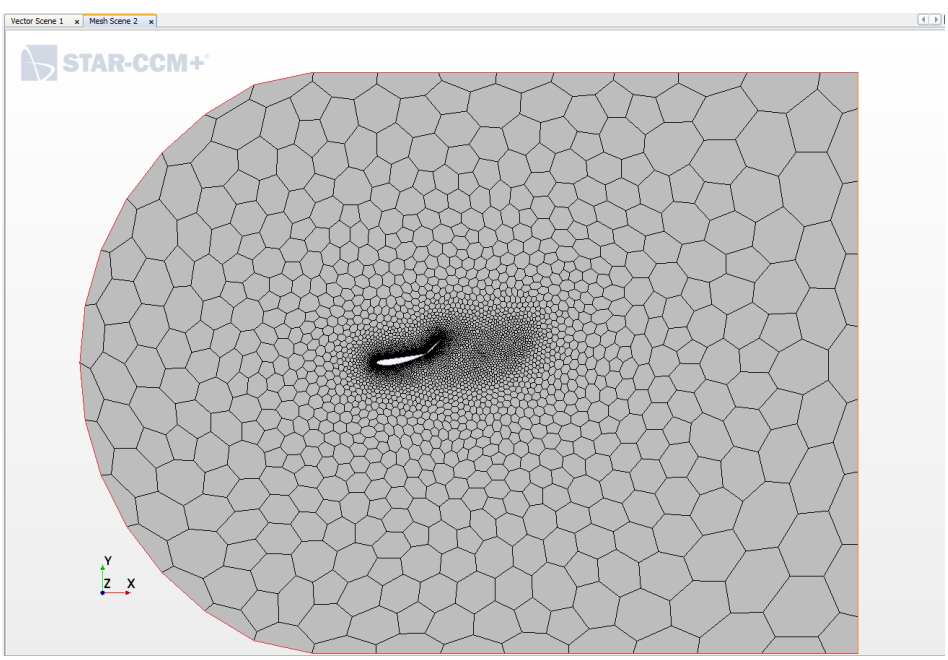
Objective

To evaluate the aerodynamic performance of a multi-element rear wing using CFD simulations, focusing on lift coefficient (Cl), drag coefficient (Cd), and aerodynamic efficiency (|Cl|/Cd), and to compare turbulence models.

Methodology

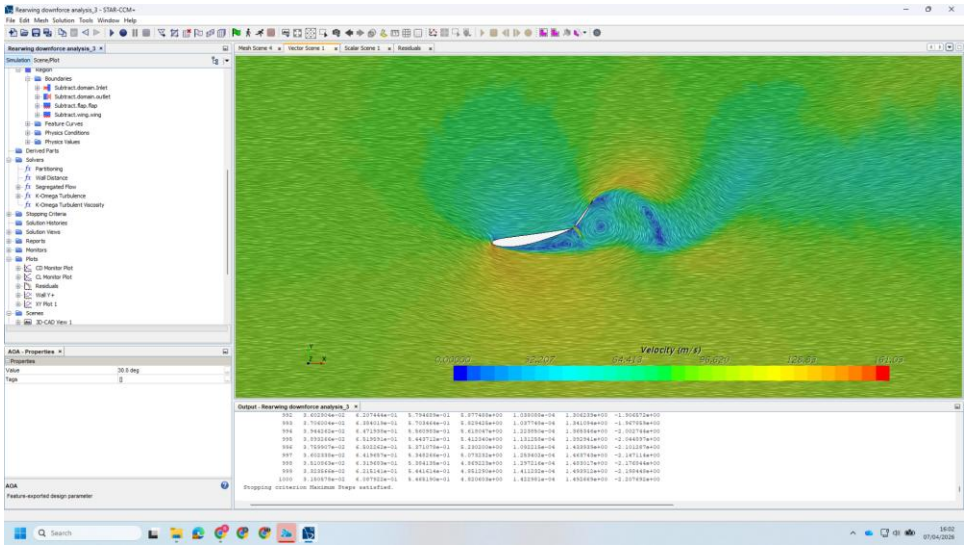
- Software: STAR-CCM+
- Models: Spalart–Allmaras (SA), k-ω SST
- Flow: Incompressible, steady RANS
- Wall treatment: Low y+ (≈1)
- Mesh: Prism layers with refinement in slot and wake
- Angles of attack analysed: 0°–45°

Mesh

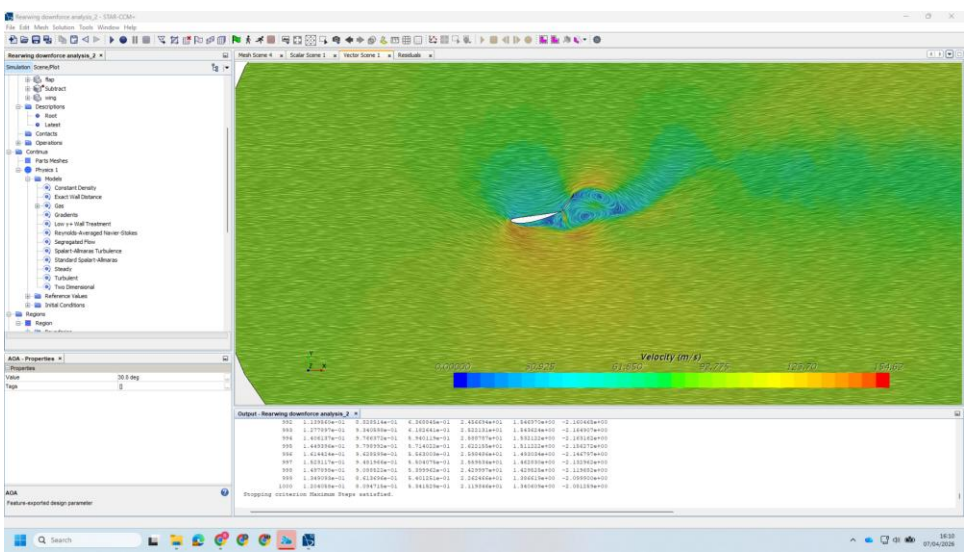


Flow Field Comparison (30° Angle of Attack)

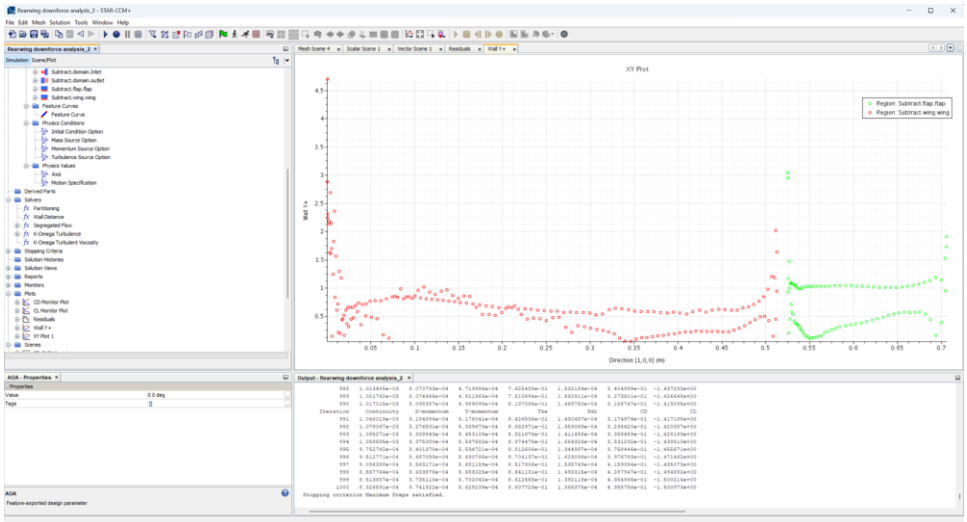
k-ω SST Model:



Spalart–Allmaras Model:



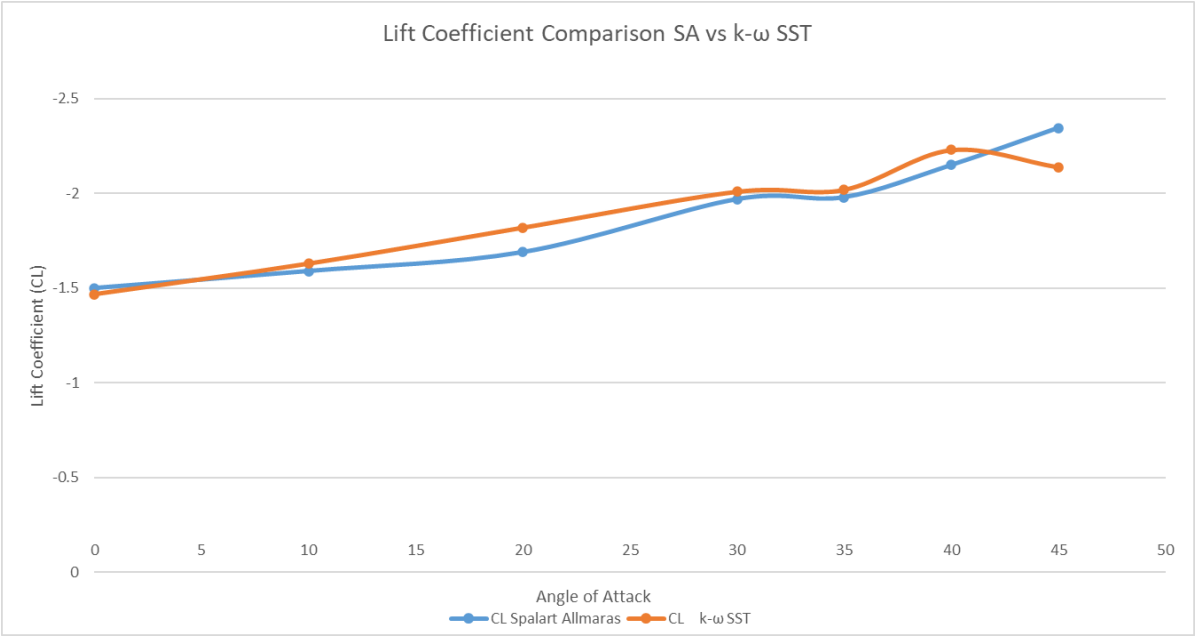
Wall y+ Distribution

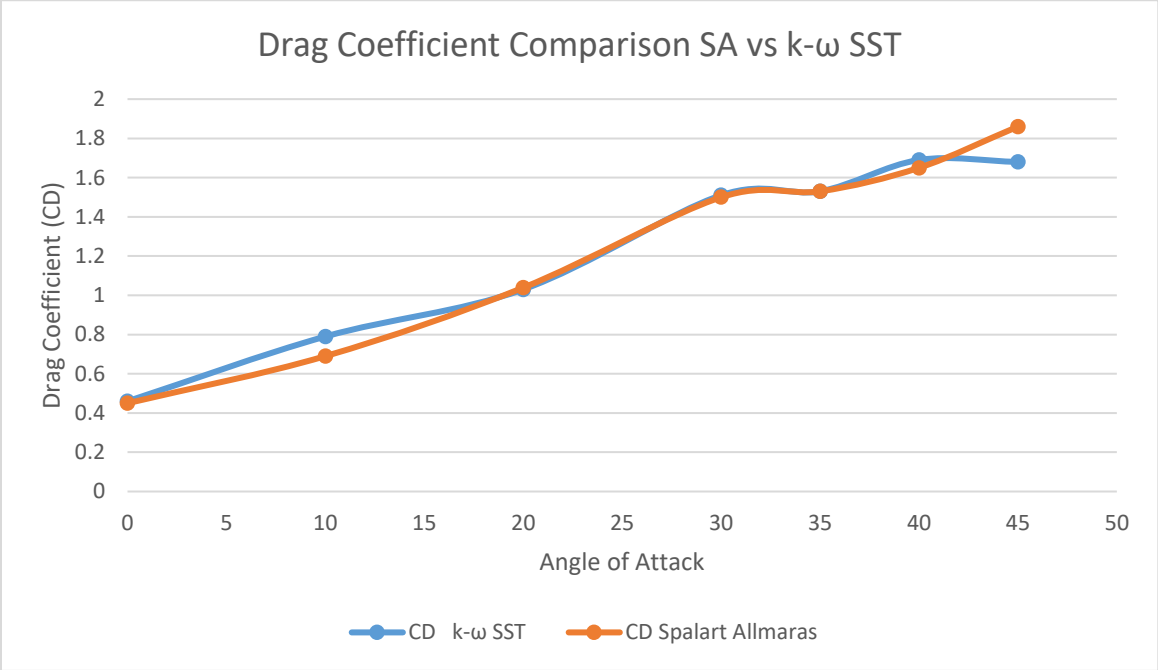


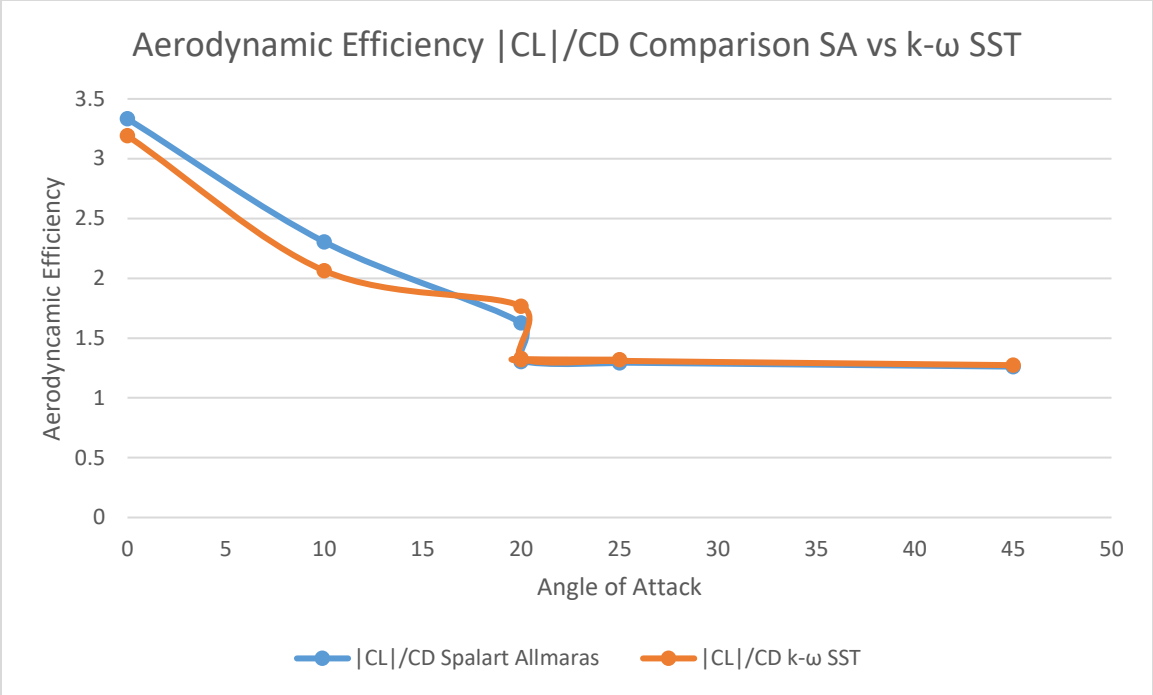
Turbulence Model Comparison

The k-ω SST model predicts higher drag compared to Spalart–Allmaras, particularly at higher angles of attack, due to improved modelling of flow separation. While both models agree at low angles, divergence increases as adverse pressure gradients develop. This results in lower aerodynamic efficiency (|Cl|/Cd) predicted by SST, indicating more realistic behaviour in separated flow conditions.

Angle fo attack	CL Spalart Allmaras	CL k-ω SST	CD k-ω SST	CD Spalart Allmaras	|CL|/CD Spalart Allmaras	|CL|/CD k-ω SST
0	-1.5	-1.468	0.46	0.45	3.333333333	3.191304348
10	-1.59	-1.63	0.79	0.69	2.304347826	2.063291139
20	-1.69	-1.82	1.03	1.039	1.626564004	1.766990291
30	-1.97	-2.01	1.51	1.5	1.313333333	1.331125828
35	-1.98	-2.02	1.53	1.53	1.294117647	1.320261438
40	-2.15	-2.23	1.69	1.65	1.303030303	1.319526627
45	-2.3463	-2.14	1.68	1.86	1.261451613	1.273809524


Lift Coefficient Comparison SA vs k-ω SST
-2.5
-2
-1.5
-1
-0.5
0
Lift Coefficient (CL)
0
5
10
15
20
25
30
35
40
45
50
Angle of Attack
CL Spalart Allmaras
CL k-ω SST


Drag Coefficient Comparison SA vs k-ω SST
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
Drag Coefficient (CD)
0
5
10
15
20
25
30
35
40
45
50
Angle of Attack
CD k-ω SST
CD Spalart Allmaras



Stall Behaviour Analysis

The Cl vs angle of attack trend shows a steady increase in downforce (negative Cl) up to approximately 40°–45°, after which the curve flattens, indicating the onset of stall. Unlike a single airfoil, the stall is gradual due to the multi-element configuration, where slot flow energises the boundary layer and delays separation. Beyond this region, separation becomes dominant, resulting in diminishing gains in downforce and increased drag.

Aerodynamic Efficiency Analysis

The aerodynamic efficiency (|Cl|/Cd) is highest at low angles of attack, where the flow remains attached and drag is minimal. As the angle increases, drag rises rapidly due to increasing adverse pressure gradients and flow separation, leading to a decrease in efficiency. The optimal operating range is identified between 10°–20°, where a balance between downforce and drag is achieved. At higher angles, although downforce increases, efficiency drops significantly, making these conditions less practical.

Conclusion

The k-ω SST model provides more realistic predictions for separated flow and is better suited for validation studies. Spalart–Allmaras is computationally efficient and suitable for initial design

exploration. The rear wing demonstrates a balance between efficiency and performance at moderate angles, with stall occurring gradually due to multi-element interaction.